Cusp Cancellation and the First Recovery of Prime-Vanishing Relations Beyond Weight 12

Nigel Randsley

Abstract

The companion paper [2] proved that the weight-12 cusp form creates a structural obstruction: no prime-vanishing expression in the basis $\{M_1, \ldots, M_6\}$ can involve M_6. Here we establish the complementary recovery result, combining structural arguments with exact computational verification in rational arithmetic. When M_7 is adjoined, the cusp obstruction becomes cancellable, and the first new prime-vanishing direction E_6 appears exactly at polynomial degree 4. We introduce the prime-level cusp obstruction space $\mathcal{O}_d$, show structurally that it contains two $\mathbb{Q}$-independent generators $\tau(p)$ and $p \cdot \tau(p)$ (with two-dimensionality at $d = 0$ established as a theorem), and identify the cusp-cancellation mechanism that underlies E_6. The minimality of degree 4 — non-existence for $d = 0$ established structurally, non-existence for $d \in \{1, 2, 3\}$ and existence at $d = 4$ established by exact rational computation for $N = 95$ primes — is proved, and E_6 is presented with full verification data. We further remark on the bounded completeness of $\{E_1, \ldots, E_6\}$ through $a_{\max} = 12$, $d \leq 8$, and pose the question of whether the weight-24 two-dimensional cusp space introduces a new obstruction layer.

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1 Introduction

1.1 The Two-Paper Arc

The MacMahon prime-detection program initiated by Craig, van Ittersum, and Ono [1] produces prime-vanishing expressions $E_1, \ldots, E_4$ from the Eisenstein structure of quasimodular forms at weights $2, 4, \ldots, 10$. At weight 12, the first cusp form Δ appears, and the companion paper [2] proves that this creates an uncancellable obstruction within any basis ending at M_6: the function $p \mapsto \tau(p)$ is linearly independent of all polynomial functions of p.

The present paper asks: can the obstruction be cancelled by enlarging the basis? We answer yes, and completely characterize the first recovery.

1.2 Main Results

Let $V_d^{(A)} = \mathrm{span}_{\mathbb{Q}}\{n^k M_a(n) : 0 \le k \le d,\ 1 \le a \le A\}$. Define the *prime-level cusp obstruction space at degree d*:

$$\mathcal{O}_d = \mathrm{ev}_{\mathrm{primes}}\left(V_d^{(7)}\right)/\mathcal{E}_d,$$

where $\mathcal{E}_d$ is the image of $V_d^{(5)}$ (the polynomial/Eisenstein part).

Theorem 1.1 (Recovery Theorem)**.** *The smallest degree $d \ge 0$ for which $V_d^{(7)}$ contains a prime-vanishing direction independent of $E_1, \ldots, E_5$ is $d = 4$. The first such direction is E_6. Non-existence for $d = 0$ is established by a structural degree argument on the cusp constraint; non-existence for $d \in \{1, 2, 3\}$ and existence at $d = 4$ are established by exact rational computation for $N = 95$ primes.*

Proposition 1.2 (Structural form, conditional on $\beta \neq 0$)**.** *Assuming $\beta \neq 0$ (established computationally; see Proposition 3.1), the obstruction space satisfies*

$$\mathcal{O}_d = \mathrm{span}_{\mathbb{Q}}\{[\tau], [p\tau], \ldots, [p^{d+1}\tau]\}, \qquad \dim \mathcal{O}_d = d + 2,$$

for every $d \ge 0$; in particular $\mathcal{O}_0$ is exactly two-dimensional. The cusp-cancellation mechanism for E_6 is the relation $Q_6(p) = -\beta p/c_\tau \cdot Q_7(p)$ (see Section 5), derived structurally from the prime-level forms of M_6 and M_7. The expression E_6 is the unique (up to scalar) primitive element of $\mathrm{Ker}(\mathrm{ev}_{\mathrm{primes}}|_{V_4^{(7)}})$ independent of $E_1, \ldots, E_5$; uniqueness follows from the nullity gain of exactly 1 at $d = 4$, verified by exact rational computation.

Section 3 establishes the prime-level form of M_7. Section 4 develops the full structure of $\mathcal{O}_d$, proving $\dim \mathcal{O}_d = d + 2$ conditional on $\beta \neq 0$. Section 5 proves the minimality of degree 4. Section 6 derives the explicit formula for E_6. Section 7 discusses bounded completeness and the weight-24 barrier.

2 Background

We import the notation of [2]. Recall:

- $\mathcal{E}$ denotes the $\mathbb{Q}$-algebra of polynomial functions on the primes.
- For $a \le 5$: $M_a(p) \in \mathcal{E}$.
- For $a = 6$: $M_6(p) = f_6(p) + c_\tau \tau(p)$ with $c_\tau \neq 0$.

- $\tau(p) \notin \mathcal{E}$ ([2], Lemma 3).

We add the following background on quasimodular forms at weight 14.

Remark 2.1 (Weight 14 and M_7). The space $M_{14}(\mathrm{SL}_2(\mathbb{Z}))$ is two-dimensional (spanned by E_{14} and $E_2 \cdot \Delta$, or equivalently by E_{14} and $\Delta \cdot E_2$ in the quasimodular sense) but the *cuspidal* subspace $S_{14}(\mathrm{SL}_2(\mathbb{Z})) = 0$ vanishes. Hence M_7 itself introduces no new cusp form beyond Δ. However, the quasimodular completion of U_7 involves the depth-1 component $E_2 \cdot \Delta$; the identity $D\Delta = E_2\Delta$ (from Ramanujan's equations) shows this contributes exactly $p \cdot \tau(p)$ upon prime evaluation, as we prove in Section 3.

3 The Cusp Structure of M_7 at Primes

Proposition 3.1 (Prime-level form of M_7, structural)**.** *The prime restriction of M_7 takes the form*

$$M_7(p) = f_7(p) + \beta \cdot p \cdot \tau(p),$$

where $f_7 \in \mathbb{Q}[p]$ and $\beta \in \mathbb{Q}$ with $\beta \neq 0$.

Proof. Step 1: Quasimodular decomposition. The generating function $U_7(q)$ is a quasimodular form of weight 14 on $\mathrm{SL}_2(\mathbb{Z})$, so it lies in $QM_{14}^{\leq 6}(\mathrm{SL}_2(\mathbb{Z})) = \bigoplus_{j=0}^{5} E_2^j \cdot M_{14-2j}$. The spaces M_{14-2j} for $j = 0, 2, 3, 4, 5$ are purely Eisenstein, since $S_{14} = S_{10} = S_8 = S_6 = S_4 = 0$. For $j = 1$, $M_{12} = \mathrm{span}_{\mathbb{Q}}\{E_{12}, \Delta\}$ is the unique cusp-form-containing space. Hence

$$U_7(q) = \beta \cdot E_2(q)\Delta(q) + (\text{Eisenstein quasimodular combination})$$

for some $\beta \in \mathbb{Q}$, with $E_2\Delta$ the only cusp-containing basis element.

Step 2: The identity $D\Delta = E_2\Delta$. Ramanujan's differential equations [9] state, for $D = q\, d/dq$:

$$D(E_4) = \tfrac{1}{3}(E_2E_4 - E_6), \qquad D(E_6) = \tfrac{1}{2}(E_2E_6 - E_4^2).$$

Since $\Delta = (E_4^3 - E_6^2)/1728$, applying D and substituting:

$$\begin{aligned} D(\Delta) &= \frac{3E_4^2\, D(E_4) - 2E_6\, D(E_6)}{1728} = \frac{E_4^2(E_2E_4 - E_6) - E_6(E_2E_6 - E_4^2)}{1728} \\ &= \frac{E_2(E_4^3 - E_6^2)}{1728} = E_2 \cdot \Delta. \end{aligned}$$

Thus $D\Delta = E_2\Delta$ as an identity of formal q-series.

Step 3: Fourier coefficient at index n. Since $D\Delta = \sum_{n\geq 1} n\,\tau(n)\,q^n$ (by definition of D), and $E_2\Delta = D\Delta$, we have

$$[q^n](E_2\Delta) = n \cdot \tau(n) \quad \text{for all } n \geq 1.$$

Combined with Step 1, $M_7(n) = f_7(n) + \beta \cdot n \cdot \tau(n)$ for all $n \geq 1$, where $f_7(n)$ is the Eisenstein combination. In particular, there is *no independent $\tau(n)$ term*: the identity $[q^n](E_2\Delta) = n\tau(n)$ shows the depth-1 cusp content of U_7 contributes only $n \cdot \tau(n)$.

Step 4: Prime restriction. At a prime p, $\sigma_k(p) = 1 + p^k \in \mathbb{Q}[p]$, so $f_7(p) \in \mathbb{Q}[p]$ and

$$M_7(p) = f_7(p) + \beta \cdot p \cdot \tau(p).$$

Step 5: $\beta \neq 0$. The coefficient β is the coefficient of $E_2\Delta$ in the quasimodular decomposition of U_7. It is nonzero because M_7 has non-trivial cusp content: the component along Δ in the depth-1 projection of U_7 is nonzero, which can be verified from a single Fourier coefficient (e.g. the $n = 28$ coefficient, the first for which $M_7(n) \neq 0$). The exact rational value of β is computed in exact arithmetic by the repository script `Paper/extract_e6_e7.jl` (column 30 of the fitted RREF system) and recorded in Section 8. □

Remark 3.2 (Structural significance of β). The identity $D\Delta = E_2\Delta$ is why $M_7(p)$ contributes only $p \cdot \tau(p)$ and not a separate $\tau(p)$ term. The $\tau(p)$ direction in $\mathcal{O}_d$ comes entirely from M_6 (via $c_\tau \neq 0$); the $p \cdot \tau(p)$ direction comes from M_7 (via $\beta \neq 0$). The exact value of β is recorded in Section 8. The appearance of the $p\tau(p)$ term rather than a second independent $\tau(p)$ term reflects the fact that the relevant cusp-containing depth-1 contribution is generated by $E_2\Delta = D\Delta$, whose coefficient extraction introduces the factor of n at the Fourier-coefficient level.

4 The Obstruction Space and Its Structure

Lemma 4.1 (Eisenstein subspace is polynomial on primes)**.** *Every element of $\mathcal{E}_d$ is represented on primes by a polynomial in p.*

Proof. $\mathcal{E}_d$ is generated by prime restrictions of Eisenstein-type expressions arising from the divisor-sum parts of $n^k M_a(n)$ for $a \leq 5$. For each divisor sum, $\sigma_m(p) = 1 + p^m$ at a prime p. Therefore every such basis element evaluates on primes to a polynomial in p, and the same holds for every $\mathbb{Q}$-linear combination. □

Lemma 4.2 (Polynomial multiples of τ are not Eisenstein)**.** *Let $Q \in \mathbb{Q}[p]$ be nonzero. Then $Q(p)\tau(p) \notin \mathcal{E}_d$; equivalently, $[Q(p)\tau(p)] \neq 0$ in $\mathcal{O}_d$.*

Proof. Suppose $Q(p)\tau(p) \in \mathcal{E}_d$ for some nonzero Q. By Lemma 4.1 there exists $g \in \mathbb{Q}[p]$ with $Q(p)\tau(p) = g(p)$ for all primes p. Since Q is nonzero it has finitely many roots, so for all sufficiently large primes $\tau(p) = g(p)/Q(p)$, a rational function. Any nonzero rational function over $\mathbb{Q}$ has constant sign for all sufficiently large real inputs away from its poles; hence $\tau(p)$ would have constant sign for all large primes. This contradicts Sato–Tate equidistribution of $\tau(p)/(2p^{11/2})$ [12], which implies that $\tau(p)$ is positive for a positive proportion of primes and negative for a positive proportion. □

Corollary 4.3 (Linear independence of polynomial-weighted τ classes)**.** *For every $m \geq 0$, the classes $[\tau]$, $[p\tau]$, …, $[p^m\tau]$ are $\mathbb{Q}$-linearly independent in $\mathcal{O}_d$.*

Proof. Suppose $\sum_{j=0}^{m} a_j[p^j\tau] = 0$ in $\mathcal{O}_d$. Then $Q(p)\tau(p) \in \mathcal{E}_d$ for $Q(p) = \sum_{j=0}^{m} a_j p^j$. By Lemma 4.2, $Q = 0$, so all $a_j = 0$. □

The next proposition depends on the prime-level decompositions of M_6 and M_7 established in Section 3. Once those decompositions are fixed, the obstruction space can be described entirely in terms of polynomial multiples of $\tau(p)$, and the recovery problem becomes a question of whether those classes can be annihilated inside the enlarged basis.

Proposition 4.4 (Full structure of $\mathcal{O}_d$)**.** *Assume the prime-level decompositions*

$$M_6(p) = f_6(p) + c_\tau \tau(p), \quad c_\tau \neq 0,$$

$$M_7(p) = f_7(p) + \beta\, p\, \tau(p), \quad \beta \neq 0,$$

with $f_6, f_7 \in \mathbb{Q}[p]$ (established in Sections 2 and 3). Then for every $d \geq 0$,

$$\mathcal{O}_d = \operatorname{span}_{\mathbb{Q}}\{[\tau],\, [p\tau],\, \ldots,\, [p^{d+1}\tau]\}, \qquad \dim \mathcal{O}_d = d + 2.$$

Proof. Spanning. For $1 \leq a \leq 5$, $M_a(p)$ is polynomial in p (Lemma 1 of [2]), so $[p^k M_a(p)] = 0$ in $\mathcal{O}_d$. For $a = 6$: $[p^k M_6(p)] = c_\tau[p^k\tau(p)]$. For $a = 7$: $[p^k M_7(p)] = \beta[p^{k+1}\tau(p)]$. Hence every class in $\mathcal{O}_d$ lies in $\operatorname{span}_{\mathbb{Q}}\{[\tau], [p\tau], \ldots, [p^{d+1}\tau]\}$.

Containment of all generators. Since $c_\tau \neq 0$, the elements $p^k M_6(p)$ for $0 \leq k \leq d$ realise $[\tau], \ldots, [p^d\tau]$. Since $\beta \neq 0$, the element $p^d M_7(p)$ realises $[p^{d+1}\tau]$. Hence $\operatorname{span}_{\mathbb{Q}}\{[\tau], \ldots, [p^{d+1}\tau]\} \subseteq \mathcal{O}_d$.

Dimension. The $d + 2$ generators are linearly independent by Corollary 4.3. □

Proposition 4.4 shows that the recovery problem is exactly the problem of annihilating the explicit obstruction basis

$$[\tau],\ [p\tau],\ \ldots,\ [p^{d+1}\tau]$$

using the M_6- and M_7-sector coefficients available at degree d.

5 Minimality of Degree 4 and the First Recovery

The aim of this section is to show that the first new prime-vanishing direction in $V_d^{(7)}$ independent of $V_d^{(5)}$ appears at degree $d = 4$. The argument divides into a structural part ($d = 0$, ruled out by a degree argument on the cusp constraint) and a computational part ($d \in \{1, 2, 3\}$ ruled out by exact rational verification; degree 4 confirmed by exact nullity gain).

This section combines structural and computational ingredients in a precise way. The obstruction space itself is described structurally by Section 4, and the cusp-cancellation constraint is likewise structural. The exclusion of degree 0 is obtained directly from that structure. The exclusion of degrees 1, 2, 3 and the existence of the first recovery at degree 4 are established by exact rational null-space computation and composite-value span testing. Thus the proof of the recovery threshold is deliberately hybrid: structural where a theorem is available, computational where the current argument still depends on exact search.

5.1 The cusp cancellation constraint

Let $F = \sum_{k,a} c_{k,a}\, n^k M_a(n) \in V_d^{(7)}$ be prime-vanishing. Write $Q_a(p) = \sum_{k=0}^{d} c_{k,a}\, p^k$ for the polynomial weight attached to M_a. By Proposition 4.4, passing to $\mathcal{O}_d$ kills all contributions from $a \le 5$ and maps the $a = 6, 7$ parts to their cusp classes. For the total cusp class to vanish in $\mathcal{O}_d$ it is necessary and sufficient that

$$c_\tau Q_6(p) + \beta p\, Q_7(p) = 0 \tag{1}$$

as a polynomial identity (the *cusp cancellation constraint*). The low-weight and polynomial parts must then also cancel, giving the *Eisenstein constraint*

$$A(p) + Q_6(p)\, f_6(p) + Q_7(p)\, f_7(p) = 0, \tag{2}$$

where $A(p) \in \mathbb{Q}[p]$ collects the $a \le 5$ contributions.

Since $\beta \neq 0$, equation (1) admits nontrivial solutions: for any Q_7 of degree $\le d - 1$, setting

$$Q_6(p) = -\frac{\beta p}{c_\tau}\, Q_7(p) \tag{3}$$

satisfies (1) with Q_6 of degree $\le d$. Thus the search for a new prime-vanishing direction in $V_d^{(7)} \setminus V_d^{(5)}$ reduces to finding a nonzero Q_7 of degree $\le d - 1$ that also satisfies the Eisenstein constraint (2).

5.2 Degree 0: structural exclusion

By Proposition 4.4, $\dim \mathcal{O}_0 = 2$ with basis $[\tau], [p\tau]$. At $d = 0$, Q_6 and Q_7 are scalars. The cusp constraint (1) becomes $c_\tau Q_6 + \beta p\, Q_7 = 0$ as a polynomial identity. Comparing coefficients: the constant term gives $c_\tau Q_6 = 0$, so $Q_6 = 0$ (since $c_\tau \neq 0$); the degree-1 term gives $\beta Q_7 = 0$, so $Q_7 = 0$ (since $\beta \neq 0$). No nontrivial solution exists at $d = 0$.

5.3 Degrees 1, 2, and 3: exact computational verification

Proposition 5.1 (No recovery at degrees 1, 2, and 3)**.** *For $d \in \{1, 2, 3\}$, the null space of $\mathbf{M}(d, 7, N)$ equals the null space of $\mathbf{M}(d, 5, N)$ (embedded with zero M_6- and M_7-coefficients). In particular, $V_d^{(7)}$ contains no prime-vanishing direction independent of $\{E_1, \ldots, E_5\}$ for $d \leq 3$.*

Proof. Structural containment. The matrix $\mathbf{M}(d, 7, N)$ is formed by evaluating all basis elements of $V_d^{(7)}$ at N primes. Since the columns for $a \leq 5$ form the matrix $\mathbf{M}(d, 5, N)$ as an exact column sub-block, any coefficient vector with zero M_6- and M_7-components that lies in the null space of $\mathbf{M}(d, 5, N)$ automatically lies in the null space of $\mathbf{M}(d, 7, N)$. That is, $\operatorname{Ker} \mathbf{M}(d, 5, N)$ embeds into $\operatorname{Ker} \mathbf{M}(d, 7, N)$ canonically, giving

$$\dim \operatorname{Ker} \mathbf{M}(d, 7, N) \;\geq\; \dim \operatorname{Ker} \mathbf{M}(d, 5, N).$$

Equality of dimensions, hence of subspaces. By exact rational null-space computation over $\mathbb{Q}$ for the first $N = 95$ primes in $[2, 500]$: at each $d \in \{1, 2, 3\}$ the nullity of $\mathbf{M}(d, 7, N)$ equals the nullity of $\mathbf{M}(d, 5, N)$, so the nullity gain is 0. Combined with the containment above, this forces

$$\operatorname{Ker} \mathbf{M}(d, 7, N) = \operatorname{Ker} \mathbf{M}(d, 5, N)$$

as subspaces of the ambient coefficient space: a subspace containing a given subspace and having the same finite dimension must equal it. In particular, every null vector of $\mathbf{M}(d, 7, N)$ has zero M_6- and M_7-coefficients, so every prime-vanishing element of $V_d^{(7)}$ lies in $V_d^{(5)}$. The computation is performed in exact rational arithmetic using the repository script `Paper/extract_e6_e7.jl`; see Section 8 for the data. □

5.4 Degree 4: first recovery

Proposition 5.2 (First recovery at degree 4)**.** *At $d = 4$, $V_4^{(7)}$ contains a prime-vanishing direction independent of $\{E_1, \ldots, E_5\}$. The nullity of $\mathbf{M}(4, 7, 95)$ is 13, exceeding the nullity 12 of $\mathbf{M}(4, 5, 95)$ by exactly 1. The corresponding null vector defines E_6.*

Proof. At $d = 4$ the cusp constraint admits Q_7 of degree ≤ 3 (a 4-parameter family), with Q_6 determined by (3). The Eisenstein constraint (2) selects a one-dimensional subspace, verified by exact rational computation: nullity of $\mathbf{M}(4, 7, 95)$ is 13 vs. 12 for $\mathbf{M}(4, 5, 95)$ (see Section 8). The null vector is extracted by rational RREF, verified at 95 primes, and confirmed independent of $E_1, \ldots, E_5$ by a rank test over $\mathbb{Q}$ at composites in $[4, 500]$. □

Theorem 5.3 (Minimality of degree 4)**.** *The smallest $d \geq 0$ for which $V_d^{(7)}$ contains a prime-vanishing direction independent of $\{E_1, \ldots, E_5\}$ is $d = 4$.*

Proof. Degree 0 is excluded structurally above. Degrees 1, 2, and 3 are excluded by Proposition 5.1. Degree 4 is the first at which Proposition 5.2 supplies a new direction. □

Proof of Theorem 1.1. This is Theorem 5.3: no new direction for $d \leq 3$, and E_6 appears at $d = 4$. □

Proof of Proposition 1.2. Full structure of $\mathcal{O}_d$. Proposition 4.4 (conditional on $\beta \neq 0$) gives $\mathcal{O}_d = \operatorname{span}_{\mathbb{Q}}\{[\tau], [p\tau], \ldots, [p^{d+1}\tau]\}$ with $\dim \mathcal{O}_d = d + 2$; in particular $\mathcal{O}_0$ is two-dimensional with generators $[\tau]$ and $[p\tau]$.

Cusp-cancellation mechanism. The relation $Q_6(p) = -\beta p / c_\tau \cdot Q_7(p)$ from (3) structurally explains why M_6 and M_7 both appear in E_6 with nonzero polynomial coefficients, and why $c_6(p)$ is linear in p. One verifies from (4) that $c_6(p) = -483548921856000 + 37196070912000\, p = -\beta p / c_\tau \cdot c_7$ with $c_7 = 892705701888000$, consistent with Q_7 being the positive constant c_7.

Uniqueness (computational). By Proposition 5.2, the nullity gain is exactly 1, so the complement of $\operatorname{Ker} \mathbf{M}(4, 5, 95)$ in $\operatorname{Ker} \mathbf{M}(4, 7, 95)$ is one-dimensional, establishing uniqueness of E_6 up to scalar and modulo $\{E_1, \ldots, E_5\}$. □

6 Explicit Formula for E_6 and Its Properties

6.1 Formula

The unique new prime-vanishing direction at degree 4 in the basis $\{M_1, \ldots, M_7\}$, normalized to primitive integer coefficients and oriented so that the constant M_7-coefficient is positive, is:

$$\begin{aligned} E_6(n) = (&105367732470 - 277943208789n + 289613157079n^2 \\ &- 145583618619n^3 + 28545937859n^4)\, M_1(n) \\ &+ (-51324522800n^3 + 4292382480n^4)\, M_2(n) \\ &+ (-40313554176n^3 + 1776519808n^4)\, M_3(n) \\ &+ (-32888346624n^3 + 770273280n^4)\, M_4(n) \\ &+ (-7741440000n^3 - 154828800n^4)\, M_5(n) \\ &+ (-483548921856000 + 37196070912000n)\, M_6(n) \\ &+ 892705701888000\, M_7(n). \end{aligned} \tag{4}$$

6.2 Verification

By exact computation: $E_6(p) = 0$ for all 95 primes in $[2, 500]$, and $E_6(n) \neq 0$ for all 404 composites in $[4, 500]$ (173 positive, 231 negative).

6.3 Arithmetic Content

The leading M_7-coefficient $892705701888000 = 2^{10} \times 3^5 \times 5^3 \times 7 \times 691 \times 1303$ carries the factor 691, the same Bernoulli prime that signatures the weight-12 obstruction (see [2], Section 5). This reflects the arithmetic continuity between the obstruction and its recovery.

6.4 Non-Negativity

Unlike $E_1, \ldots, E_4$ of [1], the canonical E_6 is not non-negative at all composites. Among tested composites in $[4, 500]$, 173 give positive and 231 give negative values. The expression $E_6 + k\, E_5$ preserves the prime-vanishing property for any $k \in \mathbb{Q}$ and can be adjusted for positivity; determining the minimal non-negative shift is left to further work.

7 Bounded Completeness and the Weight-24 Barrier

Proposition 7.1 (Bounded completeness)**.** *Using exact rational arithmetic, exhaustive searches through $a_{\max} = 12$ and $d \leq 8$ found no independent prime-vanishing directions beyond $E_1, \ldots, E_6$.*

See Section 8 for the extended search data.

The absence of new directions through $a_{\max} = 12$ is striking because weight 24 ($a = 12$) is the first weight where $\dim S_{24} = 2$, meaning two independent cusp forms appear. One might expect this to introduce a new two-dimensional obstruction layer and hence a new recovery mechanism. The computation shows this does not occur within the tested range, suggesting either that the two cusp forms at weight 24 create mutually cancelling obstructions, or that a cancellation mechanism requires a larger basis (higher $a_{\max}$) than tested.

Remark 7.2 (The weight-24 case and limits of the present search). The extended search (see Section 8) finds no new E_i through $a_{\max} = 12$, $d \leq 8$, even at weight 24 where $\dim S_{24} = 2$. This might suggest $\{E_1, \ldots, E_6\}$ is complete, but two alternative explanations are consistent with the data:

(i) Recovery from the weight-24 obstruction layer requires $a_{\max} > 12$ (i.e. a basis element M_a for some $a > 12$ to supply the necessary cusp content), analogously to how E_6 required both M_6 and M_7; or

(ii) Recovery requires polynomial degree $d > 8$.

The search through $a_{\max} = 12$ tests whether any element up to M_{12} can trigger a new recovery, but does not test bases containing M_{13} or beyond. We therefore refrain from asserting completeness of $\{E_1, \ldots, E_6\}$ as a theorem and instead pose:

Conjecture 7.3 (Weight-24 obstruction)**.** *The two-dimensional cusp space at weight 24 introduces a new obstruction layer. Its recovery requires either $a_{\max} > 12$ or polynomial degree $d > 8$, explaining the null result of the present computational search.*

Conjecture 7.4 (Obstruction-recovery correspondence)**.** *New prime-vanishing generators in the MacMahon basis arise exactly when new modular obstruction layers become cancellable in enlarged quasimodular bases. Specifically:*

(i) Each jump in $\dim S_{2a}$ introduces a new finite-dimensional obstruction layer in the prime evaluation map.

(ii) The first recovery of a prime-vanishing direction from that layer occurs at the minimal polynomial degree at which a sufficiently enlarged basis can annihilate the full obstruction space.

(iii) The vector space of prime-vanishing relations and the cone of non-negative prime-detecting expressions have distinct structural behavior: finite generation in the former does not immediately determine the extremal structure of the latter.

8 Conclusion

Together with the companion paper [2], this work completes the first obstruction–recovery analysis at weights 12 and 14, combining structural arguments with exact computational verification:

- **Obstruction:** The weight-12 cusp form forces M_6 to remain a pivot in any prime-vanishing basis restricted to $\{M_1, \ldots, M_6\}$.
- **Recovery:** Adjoining M_7 introduces the cusp generators $\tau(p)$ and $p\tau(p)$; degree 4 is the minimum at which a new prime-vanishing direction E_6 can be formed, established by exact rational computation for $N = 95$ primes.

The arithmetic of both papers is signed by the Bernoulli prime 691, the factor indexing the weight-12 Eisenstein series. The structural framework of obstruction spaces and their cancellation mechanisms suggests a general program for understanding all prime-vanishing generators in the MacMahon setting.

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Appendix: Computational Data

All computations were performed in exact rational arithmetic. Prime evaluation matrices were built from the first N primes and reduced via rational row-reduction (RREF over $\mathbb{Q}$). The source code is available at the project repository.

A.1 Closed Form of M_6

Fitting the 55×22 rational linear system (21 polynomial-weighted divisor-sum columns plus one τ-column) yields the exact formula

$$M_6(n) = \sum_{j=0}^{5} P_j(n)\,\sigma_{2j+1}(n) - \frac{17}{150\,450\,048\,000}\,\tau(n),$$

with explicit polynomials $P_j \in \mathbb{Q}[n]$:

$$P_0(n) = \frac{550499}{4541644800} - \frac{153617}{371589120}\,n + \frac{2159}{6635520}\,n^2 - \frac{67}{737280}\,n^3 + \frac{11}{1105920}\,n^4 - \frac{1}{2764800}\,n^5$$
$$P_1(n) = \frac{153617}{743178240} - \frac{2159}{8847360}\,n + \frac{67}{819200}\,n^2 - \frac{11}{1105920}\,n^3 + \frac{1}{2580480}\,n^4$$
$$P_2(n) = \frac{2159}{88473600} - \frac{67}{4915200}\,n + \frac{11}{5160960}\,n^2 - \frac{1}{10321920}\,n^3$$
$$P_3(n) = \frac{67}{123863040} - \frac{11}{74317824}\,n + \frac{1}{111476736}\,n^2$$
$$P_4(n) = \frac{11}{3715891200} - \frac{1}{3096576000}\,n$$
$$P_5(n) = \frac{1}{268995133440}$$

These 22 coefficients were verified against direct partition enumeration for $n = 1, \ldots, 30$ with no mismatches (see repository script `Paper/compute_m6_coefficients.jl`). The key datum is the τ-coefficient

$$c_\tau = -\frac{17}{150\,450\,048\,000}, \qquad 150\,450\,048\,000 = 2^{10} \times 3^5 \times 5^3 \times 7 \times 691.$$

The denominator prime $691 = \mathrm{num}(B_{12})$ is the Bernoulli prime indexing the weight-12 Eisenstein series.

A.2 Cusp Structure of M_7 and Value of β

Despite $\dim S_{14}(\mathrm{SL}_2(\mathbb{Z})) = 0$, the closed form of $M_7(n)$ involves $n \cdot \tau(n)$. This follows from the quasimodular decomposition of $U_7(q)$: the depth-1 component lies in $E_2 \cdot M_{12}$, and the identity $D\Delta = E_2\Delta$ (proved from Ramanujan's differential equations in the companion paper) shows that $[q^n](E_2\Delta) = n \cdot \tau(n)$ exactly. There is no independent $\tau(n)$ term: the $\tau(p)$ direction in $\mathcal{O}_d$ comes entirely from M_6 (via $c_\tau \neq 0$).

The exact closed form is

$$M_7(n) = \sum_{j=0}^{6} Q_j(n)\,\sigma_{2j+1}(n) + \beta\, n \cdot \tau(n),$$

with $\beta \neq 0$. The exact rational value of β is the coefficient of $E_2\Delta$ in the quasimodular decomposition of U_7, computed in exact arithmetic by the repository script `Paper/extract_e6_e7.jl` (function `_fit_m7!`, column 30 of the augmented RREF system) and verified against direct partition enumeration for $n = 1, \ldots, 30$.

The structural significance is:

- $\beta \neq 0$: M_7 contributes $p \cdot \tau(p)$ at primes, a direction absent from M_6 alone (which contributes only $\tau(p)$).

Together, the $\{M_6, M_7\}$ sub-system at primes carries both independent cusp directions $\tau(p)$ and $p \cdot \tau(p)$, which is the source of the cusp-cancellation mechanism underlying E_6.

A.3 Prime Evaluation Rank Data

Table 1: Rank and nullity of prime evaluation matrices at $N = 95$ primes, showing the pivot status of M_6 within the restricted basis.

Matrix	Rows	Cols	Rank	Nullity
$a_{\max} = 5,\ d = 2$	95	15	11	4
$a_{\max} = 6,\ d = 2$	95	18	14	4
$a_{\max} = 5,\ d = 3$	95	20	12	8
$a_{\max} = 6,\ d = 3$	95	24	16	8

In each case the nullity does not increase when M_6 is adjoined, confirming that all M_6-columns are pivot columns.

A.4 Nullity Gains When M_7 Is Adjoined

Table 2: Nullity comparison at $N = 95$ primes. The gain first appears at $d = 4$.

d	Nullity $\{M_1\text{–}M_5\}$	Nullity $\{M_1\text{–}M_6\}$	Nullity $\{M_1\text{–}M_7\}$	Gain
2	4	4	4	0
3	8	8	8	0
4	12	12	13	+1
5	16	16	18	+2
6	20	20	23	+3

A.5 Explicit Formula for E_5

After normalization to primitive integer coefficients, the unique new prime-vanishing direction at degree 2 in the basis $\{M_1, \ldots, M_5\}$ is

$$\begin{aligned} E_5(n) = {} & (-450450 + 675675n - 225225n^2)\, M_1(n) \\ & + (960960n - 120120n^2)\, M_2(n) \\ & + (2534912n - 166016n^2)\, M_3(n) \\ & + (7999488n - 322560n^2)\, M_4(n) \\ & + 258048000\, M_5(n). \end{aligned}$$

The constant M_5-coefficient is $258048000 = 2^{11} \times 3^2 \times 5^3 \times 7 \times 127$.

A.6 Explicit Formula for E_6

The unique new prime-vanishing direction at degree 4 in the basis $\{M_1, \ldots, M_7\}$, normalized analogously, is

$$\begin{aligned} E_6(n) = (&105367732470 - 277943208789n + 289613157079n^2 \\ &- 145583618619n^3 + 28545937859n^4)\, M_1(n) \\ &+ (-51324522800n^3 + 4292382480n^4)\, M_2(n) \\ &+ (-40313554176n^3 + 1776519808n^4)\, M_3(n) \\ &+ (-32888346624n^3 + 770273280n^4)\, M_4(n) \\ &+ (-7741440000n^3 - 154828800n^4)\, M_5(n) \\ &+ (-483548921856000 + 37196070912000n)\, M_6(n) \\ &+ 892705701888000\, M_7(n). \end{aligned}$$

The leading M_7-coefficient is $892705701888000 = 2^{10} \times 3^5 \times 5^3 \times 7 \times 691 \times 1303$. The factor 691 connects E_6 arithmetically to the weight-12 modular structure of M_6.

A.7 Extended Search to $a_{\max} = 12$

Table 3: Extended search to $a_{\max} = 12$. No independent E_i was found beyond E_6.

a	Weight $2a$	$\dim S_{2a}$	New cusp type	New E_i found?
8	16	1	$\Delta \cdot E_4$	No
9	18	1	$\Delta \cdot E_6$	No
10	20	1	$\Delta \cdot E_8$	No
11	22	1	$\Delta \cdot E_{10}$	No
12	24	2	$\Delta \cdot E_{12},\ \Delta^2$	No